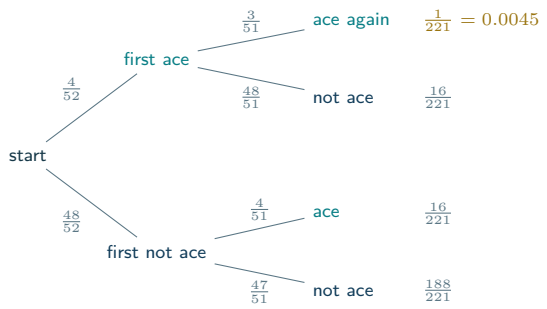
PROBABILITY LADDER 3 OF 9

Conditional probability and independence

Learning that one event happened shrinks the sample space, and independence says it leaves another event unchanged.

1 / THE RULE AND THE PICTURE

Along a branch of the tree the probabilities multiply, because each is conditional on the one before



two cards from a pack without replacement

The conditional probability of A given B restricts attention to the outcomes in B:

$$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}$$

so $P(A \text{ and } B) = P(B)\,P(A \mid B)$, and for two aces $\frac{4}{52} \cdot \frac{3}{51} = \frac{1}{221}$.

2 / SHRINKING THE SAMPLE SPACE / TWO DICE

Given that the sum is 8, only five of the 36 outcomes remain

The outcomes with sum 8 are $(2,6)$, $(3,5)$, $(4,4)$, $(5,3)$, $(6,2)$, and exactly one of them has a six on the first die, so $P(\text{first is } 6 \mid \text{sum} = 8) = \frac{1}{5}$, against the unconditional $\frac{1}{6}$. By the formula the same number is $\dfrac{1/36}{5/36} = \frac{1}{5}$, and conditioning on a sum of 7 would give $\frac{1}{6}$ again.

3 / INDEPENDENCE

Events are independent when $P(A \text{ and } B) = P(A)\,P(B)$, so that conditioning changes nothing

events A, B	$P(A \text{ and } B)$	$P(A)P(B)$	verdict
first die even, sum 7	$\frac{3}{36}$	$\frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}$	independent
first die even, sum ≥ 10	$\frac{4}{36}$	$\frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}$	dependent
sum 7, sum 8	0	$\frac{1}{6} \cdot \frac{5}{36}$	exclusive, dependent

Exclusive events with positive probabilities are never independent, since knowing one happened rules the other out.

4 / THE GAMBLER'S FALLACY / AND THE COMPLEMENT

Past tosses do not change a fair coin, and "at least one" is easiest through the complement

After five heads the next toss is still $\frac{1}{2}$ heads, although six heads in a row had probability $\frac{1}{64}$ from the start.
For at least one six in four rolls, $1 - (\frac{5}{6})^4 = 0.518$, and for at least one double six in 24 rolls of two dice, $1 - (\frac{35}{36})^{24} = 0.491$, the two numbers that separated the bets of the Chevalier de Méré.

5 / TOTAL PROBABILITY

Splitting by the first card gives the chance of the second

The second card is an ace with probability $\frac{4}{52} \cdot \frac{3}{51} + \frac{48}{52} \cdot \frac{4}{51} = \frac{12+192}{2652} = \frac{1}{13}$, the same as for the first card by symmetry. In general $P(B) = P(B \mid A)P(A) + P(B \mid \text{not } A)P(\text{not } A)$, which the next sheet turns around into Bayes' theorem.

ASIDE / THE SAME ANSWER TWICE

Counting hands with $\binom{4}{2}/\binom{52}{2}$ and multiplying along the branch both give $\frac{1}{221}$.

LINKS TO OTHER SHEETS

The counting method is in *Counting: arrangements and choices*; the next sheet is *Bayes' theorem and updating*.

Convention. $P(A \mid B)$ is the probability of A given B; cards are dealt without replacement; the dice are fair.